Hardware User Manual

AnyAIO Series

PCI-AI16HR V1.2

AJINEXTEK AXT

Product Information

Full information about other AJINEXTEK products
is available by visiting our Web Site at:
www.ajinextek.com

Useful Contact Information

Customer Support Seoul
Tel : 82-31-3426-2180 Fax: 82-31-436-2183

Customer Support Cheunan
Tel : 82-41-555-9771 Fax: 82-41-555-9773

Customer Support Daegu
Tel : 82-53-593-3700 Fax: 82-53-593-3703

AJINEXTEK's sales team is always available to assist you in making your decision the final choice of boards or systems is solely and wholly theresponsibility of the buyer. AJINEXTEK's entire liability in respect of the board or systems is as set out in AJINEXTEK's standard terms and conditions of sale

ⓒ Copyright 2007AJNEXTEK co.ltd. All rights reserved.

Contents

1. Overview _____ 5

1.1. Foreword 5

1.2. Application 6

2. Features _____ 7

2.1. Specification 7

3. Installation _____ 9

3.1. Hardware Installation 9

3.2. Software Installation 9

4. Configuration _____ 10

4.1. Block Diagram of PCI-AI16HR Board 10

4.2. Hardware Description 12

4.2.1. Major parts of PCI-AI16HR board 12

4.2.2. LED status check 14

4.2.3. External Connector Pin Array 15

4.2.4. External trigger Input stage circuit 16

4.3. Address Allocation Process 17

4.4. Register Map 18

4.4.1. Register map of the analog input block of PCI-AI16HR board 18

4.4.2. Register map of the analog output block of PCI-AI16HR board 19

5. Accessories (optional items) _____ 22

5.1. Cable (dedicated cable for PCI-AI16HR) 22

5.2. Terminal Blocks 23

5.3. Connecting Devices 25

6. Order Information (AnyBus Family) _____ 26

Revision History

Manual	PCB	Comments
Rev. 1.0 issue 1.0	Rev. 1.1	2007. 2. 26
Rev. 1.1	Rev. 1.2	2007. 11.29
Rev. 1.2	Rev. 1.2	2009. 06.15

1. Overview

1.1.Foreword

PCI-AI16HR is a half-size PC add-on card that includes 16 channels of analog input and two channels of analog output converted from digital signal with a resolution of 16 bits.

While the digital input and output channels are used to get or set the binary state (On and Off) of various devices (switches, relays, lamps,..), the analog input and output channels can be used to measure the continuous analog quantity by converting analog signal into digital value or to send the continuous analog signal to the external load by converting digital signal into analog value.

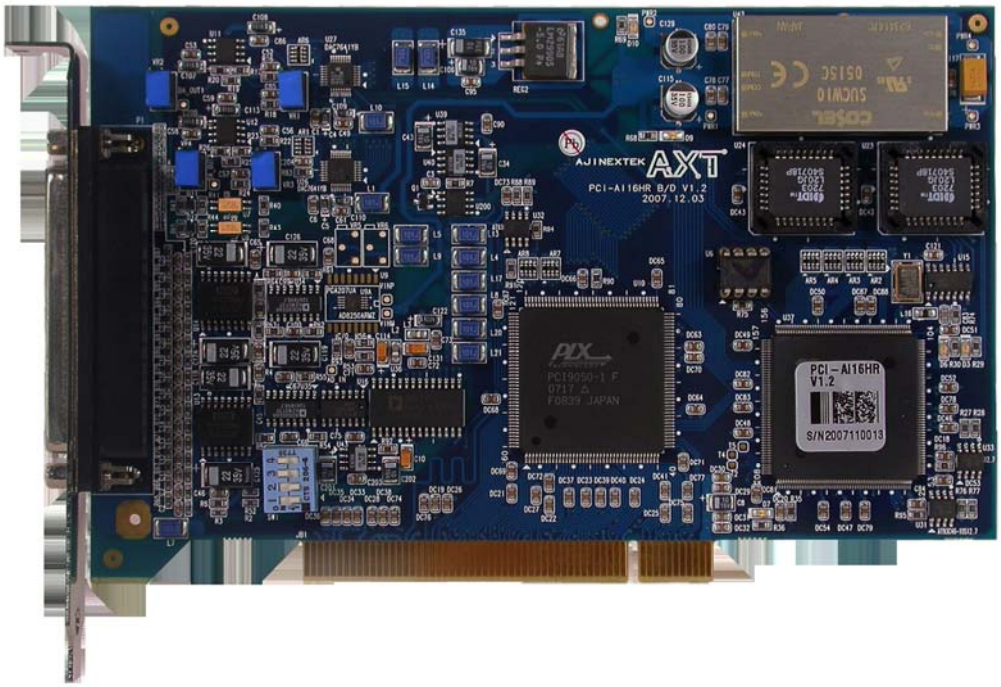
1.2.Application

PCI-AI16HR with the input and output interface to external analog signal can be used in various applications like factory automation, machine tools control, experiment and education, and so on.

- ▶ Automatic test equipment
- ▶ Equipment Interface
- ▶ Process control
- ▶ Signal Analysis
- ▶ R & D Instrumentation
- ▶ Multi-channel data acquisition and Analog voltage output
- ▶ Temperature monitoring and control
- ▶ Data acquisition for sensor devices
- ▶ And other analog signal applications

2. Features

2.1. Specification



Picture 1. Picture of PCI-AI16HR board

Table 1. PCI-AI16HR board specification

Items	PCI-AI16HR
Form Factor	Standard PCI half size(short card)
Conformance	PCI Local Bus Spec. Rev. 2.1(32bit/33㎒)
AD Converter Chip	AD976
DA conversion chip	DAC7641
Channel	Input 16 Channel, Output 2 channel
Input/Output Range	±10V
Full-Scale Error Max	±10mV
Input/Output Mode	Single-Ended - input 16ch, Output 2ch
Sampling Frequency	100Ks/sec
Resolution	16-bit
Operating Temperature Range	0 ~ +60℃

Operating Humidity Range	Below 80%
Operating Voltage	5V, Tolerance: -0.3V to 7.0V
Signal Level	CMOS logic level: High (70% ~ 100%) Low (0 ~ 20%) TTL logic level : High (2.0V) Low (0.8V)
Maximum Current Consumption	5V, 1.2A

PCI-AI16HR is an analog input and output board that contains 16 channels of single-ended analog input and two channels of analog output with a resolution of 16 bits. Without using external power supplies, internal single 5V is used to supply all required voltages to A/D and D/A converters with DC-DC converters. And the A/D conversion can be triggered by internal timer, user software, and external trigger input.

3. Installation

3.1. Hardware Installation

Inspect the contents of the package and install the board by following the procedure shown below. The product comprises PCI-AI16HR board and optional cable and terminal blocks.

① Turn off the PC and unplug the power cord.

② Determine the PCI slot (5V slot) to insert the board.

③ Remove the bracket of the selected slot.

④ In order to discharge the possible static electricity built up on the body or cloth, touch the metal part of the PC case with hands.

⑤ Insert the board to the selected PCI slot. Fix the board to the slot with the screws for this purpose.

⑥ Inspect visually whether the board is installed properly.

⑦ Plug the power cord and turn on the PC to start the system.

⑧ Check whether three power LEDs at the upper and lower part of the board are on normally.

⑨ Check whether the system boots up normally.

⑩ Check whether new hardware is found in case of Windows XP.

⑪ Install the hardware driver as below depending on the operating system.

Refer to the following installation procedure and complete the hardware installation.

Windows 2000, XP

During the hardware driver installation, specify the path (C:₩Programs Files₩EzSoftwareAXL₩INF₩AXT_PCIB.inf) of the installation file (AXT_PCIB.inf) and select the file when you are asked to enter the installation information.

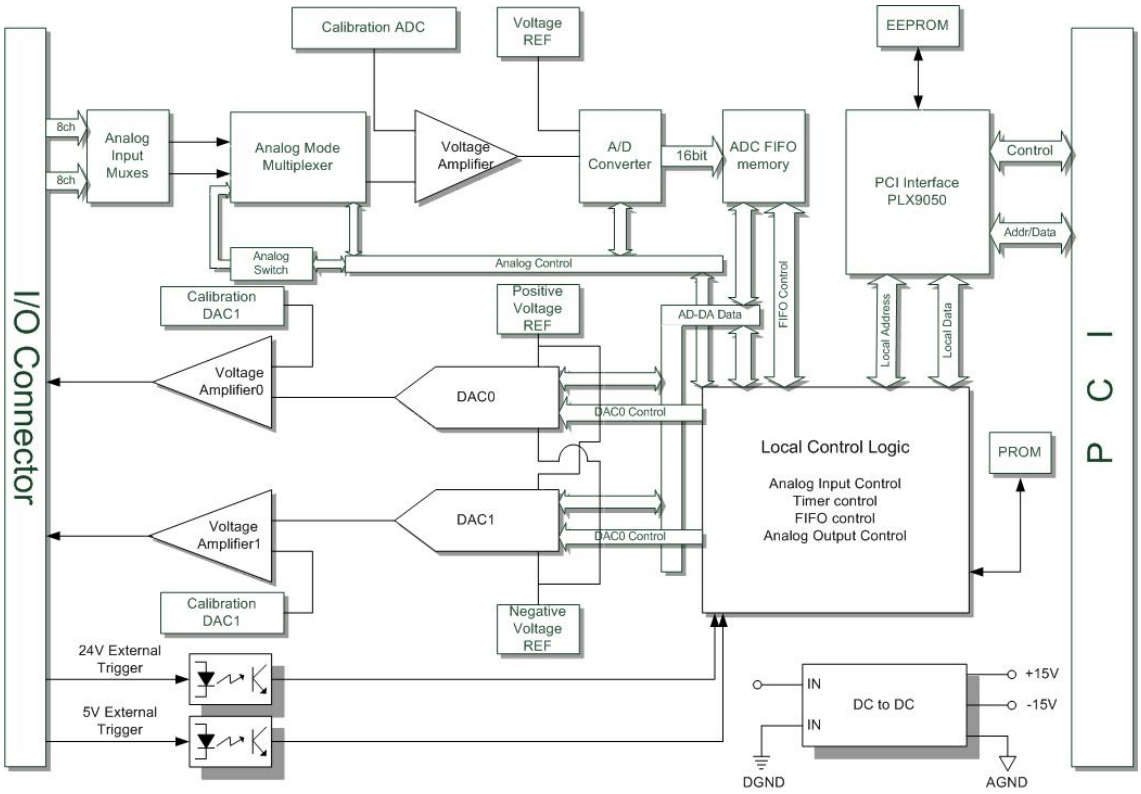
3.2. Software Installation

Refer to the EzSoftware AXL users manual and install the EzSoftware AXL software.

4. Configuration

4.1.Block Diagram of PCI-AI16HR Board

The block diagram of the PCI-AI16HR board is as shown below.



Picture 2. Block diagram of PCI-AI16HR Board

Table 2. Requirement of PCI-AI16HR board

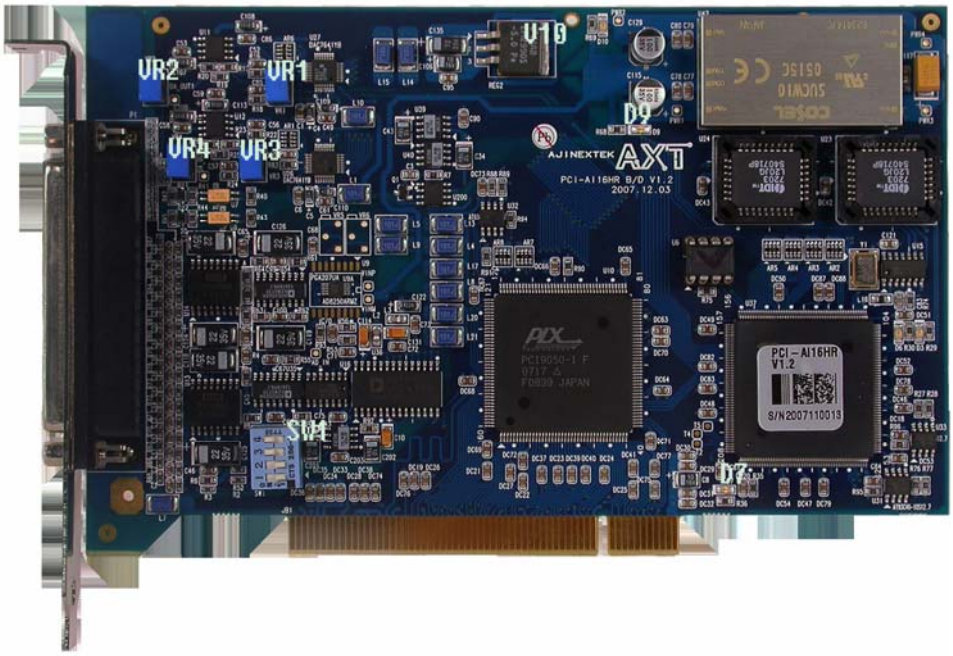
Dimension	174.63×111 ㎜
System Bus	PCI Local Bus Spec. Rev. 2.1(32bit/33㎒)
Space required	One slot (DC 5V slot)
Operating voltage	Internal voltage: DC 5V
Connector	Single column of 37 pin D-sub Female connector

As shown in the picture 2, the input signals over 16 channels are acquired in fixed time interval by the switching through 16 channels. And calibration circuit for fine tuning offset and gain is applied. In order to implement the accurate sampling period, hardware timer is implemented, and the buffer of 1K word is provided to prevent the data loss during the data transfer from the board to the PC.

And DAC device is applied on each output channel to produce analog voltage output. Reference circuit, offset adjustment circuit, and gain adjustment circuit are applied for the precision output control.

4.2.Hardware Description

4.2.1.Major parts of PCI-AI16HR board



Functions of variable resistors

VR1,3: Offset value setting for output channel 1 and 2 respectively.

VR2,4: Voltage gain setting for output channel 1 and 2 respectively

Functions of LEDs

D7,D9,D10 : Power indication of 5V, + 15V, -15V in the order of the list

Function of SW1

Functions by S1,2 settings

S1 / S2	Off	On
Off	Voltage Input DAQ	X
On	X	X

Functions by S3,4 settings

S1 / S2	Off	On
Off	X	X
On	X	Signal ended

Picture 3.Major parts of PCI-AI16HR board

Picture 3 shows the external connector of PCI-AI16HR board, offset and gain control of input voltage, offset and gain control of output voltage, power LED, and SW1 for mode configuration.

Refer to the picture 3 on the programming method of SW1. Accurate offset and gain control can be done by following the offset and gain control procedure explained below.

4.2.1.2Precision adjustment of output signal (refer to the table 3)

A. Offset adjustment - DVM(Digital Volt Meter) method - Numbers inside ( ) are for the channel 2

a Connect the pin 29(27) of the connector, output signal, to (+) signal input terminal of DVM.

b Connect the pin 28 or 30 (DA_GND) to GND of DVM.

c Execute ExConfig EzAO agent proram provided by Ajinextek.

d Set the output voltage of the channel 1(2) to 0V.

e Read the voltage with DVM.

f If the output voltage reading is 0V, then terminated the process, continue otherwise.

g Adjust the variable resistor VR1 (3) shown in the picture 4 by one turn to the left or right.

h Read the output voltage, and repeat the step g. until there is a change in the reading, and continue to the next step.

i If the voltage reading is higher than the previous value, turn the control in opposite direction.

j Read the output voltage.

k If the value is not 0V, repeat the steps i. - k., and terminate the process when the voltage reading becomes 0V.

B. Gain adjustment - DVM method - Numbers in ( ) are for the channel 2

→ Gain adjustment procedure is similar to offset adjustment described above, and the procedure is as followings.

a Connect the pin 29(27) of the connector, output signal, to (+) signal input terminal of DVM.

b Connect the pin 28 or 30 (DA_GND) to GND of DVM.

c Execute ExConfig EzAO agent program provided by Ajinextek.

d Set the output voltage of the channel 1(2) to 9.9V.

e Read the voltage with DVM.

f If the output voltage reading is not 9.9V, execute the process below.

g Adjust the variable resistor VR2 (4) shown in the picture 4 by one turn to the left or right.

h Read the current average input voltage, and repeat the step g. until there is a change in the reading. Continue to the next step when there is a change in the reading.

i If the read voltage(absolute value) is higher than the previous value, turn the control in opposite direction.

j Read the output voltage.

k If the value is not 9.9V, repeat the steps i. - k., and terminate the process when the voltage reading becomes 9.9V.

4.2.1.2 Precision adjustment of the input signal(refer to the table 3)

Offset and gain control of the analog input signal is automatically adjusted when the system is restarted after the analog output offset and again control adjustments are done and EzSoftwareAXL is terminated. Analog inputs are designed to be calibrated automatically when the board is initialized.

4.2.2.LED status check

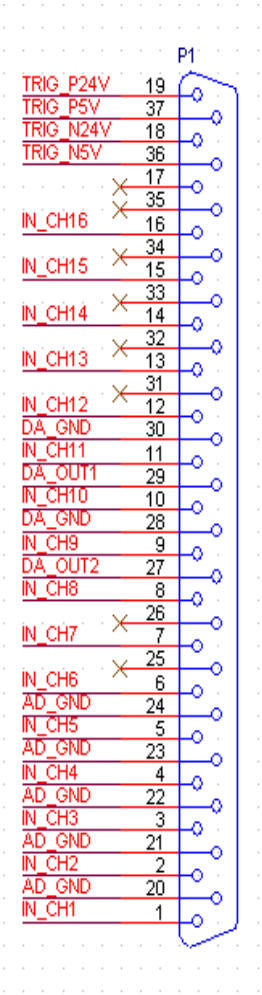
Power LEDs are for checking the power supply status of the board, and refer to the picture 3 for the functions of each LED.

4.2.3.External Connector Pin Array

External connector of PCI-AI16HR is D-SUB 37 pin. Pin definitions of the external connector are as shown in the table 4 below.

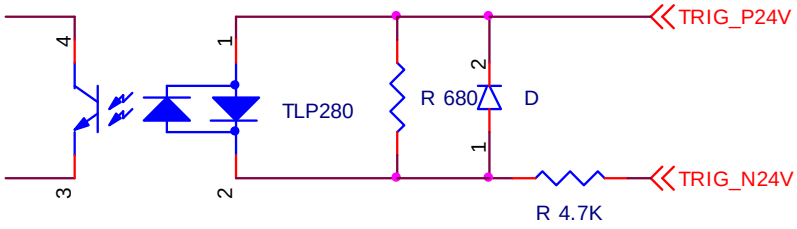
Table 3. Pin definitions of the external connectors of PCI-AI16HR board

Input		
Name	**Pin Number**	**Description**
IN_CH1	1	Analog Voltage Input ch1
IN_CH2	2	Analog Voltage Input ch2
IN_CH3	3	Analog Voltage Input ch3
IN_CH4	4	Analog Voltage Input ch4
IN_CH5	5	Analog Voltage Input ch5
IN_CH6	6	Analog Voltage Input ch6
IN_CH7	7	Analog Voltage Input ch7
IN_CH8	8	Analog Voltage Input ch8
IN_CH9	9	Analog Voltage Input ch9
IN_CH10	10	Analog Voltage Input ch10
IN_CH11	11	Analog Voltage Input ch11
IN_CH12	12	Analog Voltage Input ch12
IN_CH13	13	Analog Voltage Input ch13
IN_CH14	14	Analog Voltage Input ch14
IN_CH15	15	Analog Voltage Input ch15
IN_CH16	16	Analog Voltage Input ch16
AD_GND	20,21,22, 23,24	Analog Ground of ADC block
Output		
DA_OUT1	29	Analog Output ch1
DA_OUT2	27	Analog Output ch2
DA_GND	28,30	Analog Ground of DAC block
EXTERNAL TRIGGER		
TRIG_P24V	19	24V external trigger Positive voltage
TRIG_N24V	18	24V external trigger Negative voltage
TRIG_P5V	37	5V external trigger Positive voltage
TRIG_N5V	36	5V external trigger Negative voltage

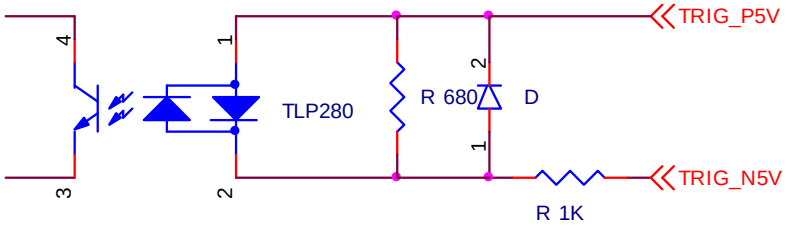


4.2.4. External trigger Input stage circuit

Picture 4 shows the circuit configuration of the external trigger. The wiring for using external trigger input signal based on the picture 4 shall be suitable to the user's trigger generation circuit. Picture 5 shows an example wiring.

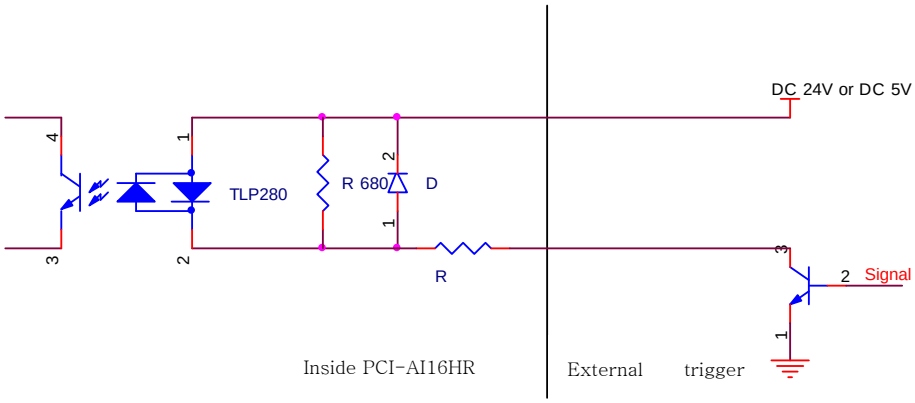


When the external input with 24V voltage level is connected



When the external input with 5V voltage level is connected

Picture 4. External Trigger Input Circuit

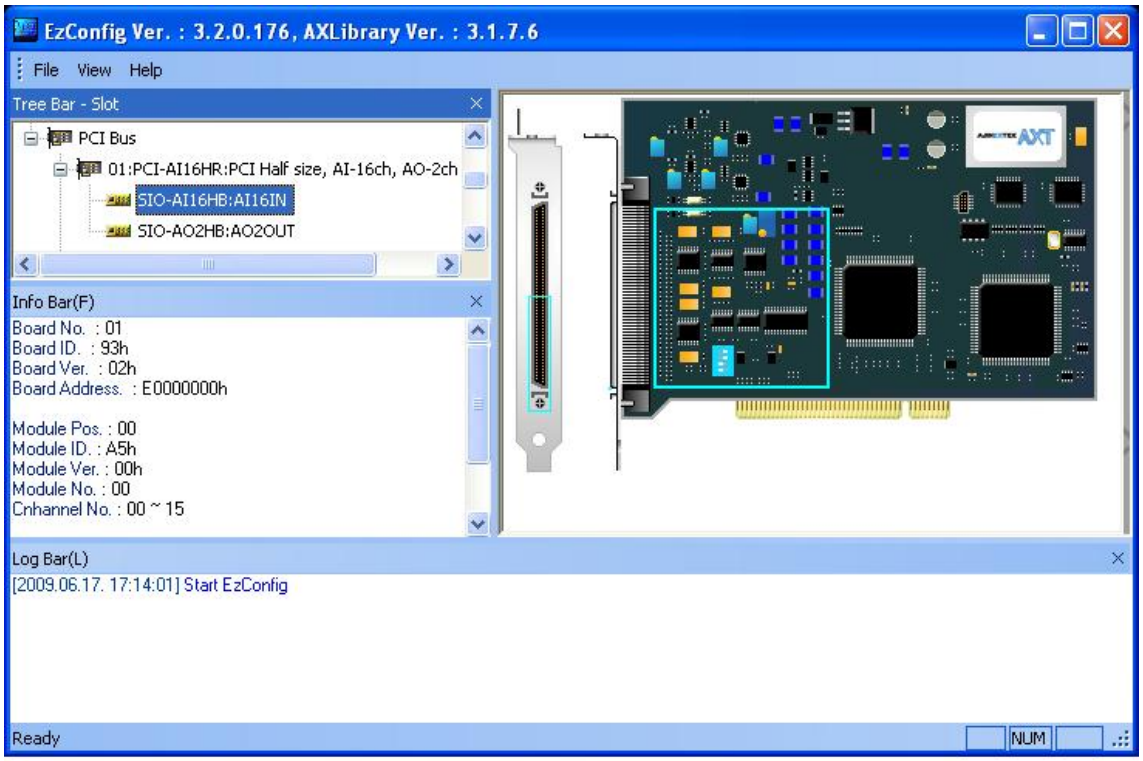


In case of open collection type of external trigger

Picture 5. External Trigger Wiring Example

4.3. Address Allocation Process

The amount of memory allocated to one PCI-AI16BR board is 64Kbye, and this memory is allocated automatically by PC ROM BIOS, and the allocation might be different depending on the used system. You can check the address easily with EzConfig program that comes with the board. And also you can check the address of the board, slot number, and used IRQ by using FindAXT.exe. Picture 6 is a screen that shows the system address used by PCI-AI16HR board, serial number, and modules installed in the sub slots using EzConfig program. You can also check the address of each installed input and output area from the information view window. The picture shown below is an example information view displaying PCI-AI16HR board by EzConfig program. You can check the address of the input and output blocks by selecting the corresponding sub-tab in the information view dialogue.



Picture 6. Information View Window of PCI-AI16HR using EzConfig Program

4.4.Register Map

4.4.1. Register map of the analog input block of PCI-AI16HR board

Table 5. Register map of the analog input block

Offset	Name	Function	Length	Initial Value
Read Port				
00h	Preamble	Preamble Register	8 bit	B6h
02h	ID	ID Register	8 bit	A5h
04h	Version	Version Register	8 bit	00h
10h	ADC Multi-channel setting register	Each bit enables or disables the A/D conversion of the corresponding channel '1' : select, '0': deselect	16 bit	0001h
12h	ADC Single-channel setting register	Register to store channel number when one channel is set to perform ADC	8 bit	0h
14h	ADC trigger control register	2To set the trigger source of ADC '000' : disable, '001' : software '010': timer '100' : external	3 bit	01h
16h	Sampling time register low word	To store low word of 32bit sampling time sampling time = time / Fclk	16 bit	0h
18h	Sampling time register high word	To store high word of 32bit sampling time	16 bit	0000h
20h	Interrupt enable register	To configure interrupt '1' : enable, '0': disable	1 bit	00h
22h	Interrupt source register	'00' : ADC done '01' : Scan end '10' : FIFO half '11' : No signal	2 bit	02h
24h	Interrupt clear	Write signal	8 bit	
26h	Interrupt flag register	Interrupt source display '00' : ADC done '01' : Scan end .10' : FIFO half '11' : No signal	2 bit	00h

30h	A/D buffer register	To store A/D converted data	16 bit	
32h	Conversion check Register	To check whether A/D conversion is complete	1bit	
36h	A/D FIFO register	Read address of the FIFO memory that stores A/D converted values	16 bit	
38h	A/D FIFO status register	Current status of FIFO memory that stores A/D converted values '001' : empty '010' : half full '100' : full	16 bit	0001h
3ah	FIFO clear	To clear FIFO memory at the write signal		
3ch	A/D software trigger	To generate the software trigger pulse at the write signal		
3eh	Output feedback S/W	High: Output CH1 feedback to IN_CH1 Low: External voltage input to IN_CH1	1bit	00h

4.4.2. Register map of the analog output block of PCI-AI16HR board

Table 6. Register map of analog output block

Offset	Name	Function	Length	Initial Value
READ PORT				
00h	Preamble	Preamble Register	8 bit	B6h
02h	ID	ID Register	8 bit	A6h
04h	Version	Version Register	8 bit	00h
10h	Ch0 read register	Read the value stored in Ch0 internal register	16 bit	
12h	Ch1 read register	Read the value stored in Ch1 internal register	16 bit	
20h	Ch0 Read register	Read the value stored in Ch0 DA register	16 bit	
22h	Ch1 Read register	Read the value stored in Ch1 DA register	16 bit	
WRITE PORT				
06h	Software reset register	Active low register to produce 0V output voltage and to clear all write registers	1 bit	0h

10h	Ch0 write register	Data is written both Ch0 internal register and DA register simultaneously, DAC produces output voltage when write is complete	16 bit	0000h
12h	Ch1 write register	Data is written both Ch1 internal register and DA register simultaneously, DAC produces output voltage when write is complete	16 bit	0000h

The timings of read and write to and from the registers in the table 5 and 6 shown above are as shown in the picture 8 and 9. The definitions of the timing related terms shown in the picture 7 and 8 are explained below

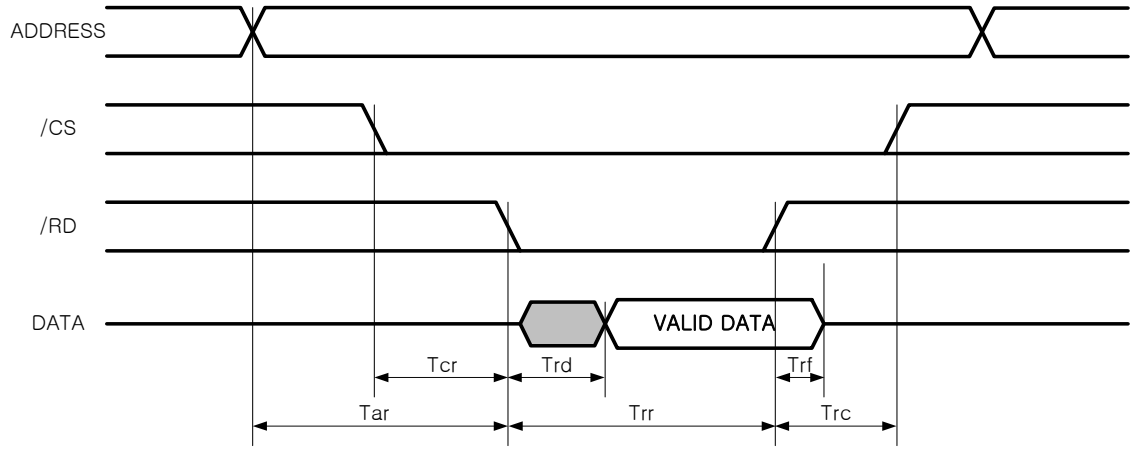
ADDRESS(LA2 ~ 16, LAD0 ~ 15): Address to access the control related internal registers of PCI-AI16HR.

/CS(/CS0): The signal to select the signal module between the input and output blocks. Input block registers can be accessed when this signal is 'LOW' and output block registers can be accessed when the level is 'HIGH'.

/RD: /RD signal goes low to read the registers of PCI-AI16HR, and the data shall be latched to the data bus of the host at the rising edge. Picture 6 below shows the timing of /RD.

/WR: /WR signal goes low to write into the registers of PCI-AI16HR.

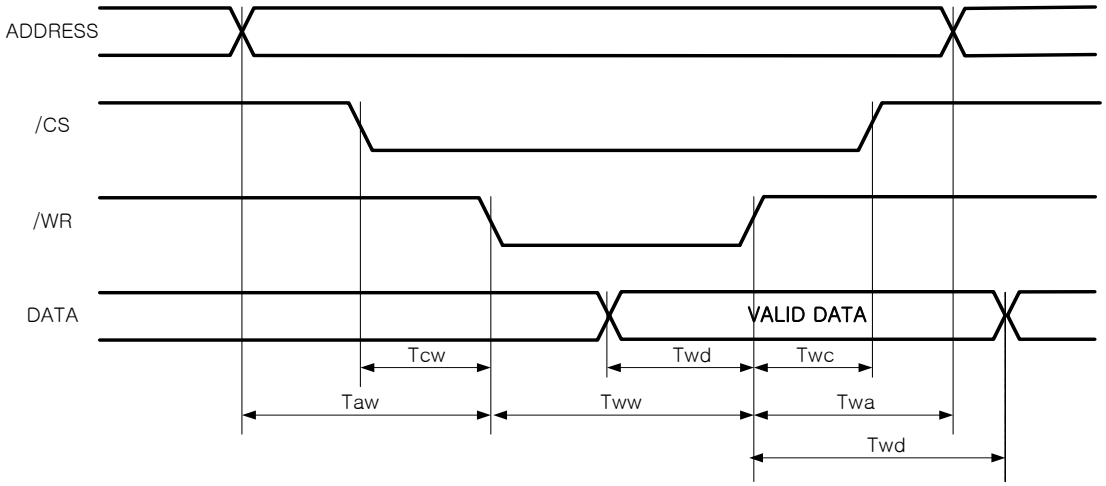
DATA (LAD0 ~ 15): Tri-sate bi-directional data bus of 16 bits connected to the data bus of the host (PC)



Picture 7. PCI-AI16HR의 Read timing diagram

Table 7. Definition of timing terms related to read timing

Symbol	Description	MIN[㎱]	MAX[㎱]
Tcr	/CS settling time(to /RD)	0	
Trd	DATA output delay time (to /RD)		40
Trc	/CS hold time to /RD	0	
Tar	ADDRESS settling time	0	
Trr	/RD pulse width	50	
Tra	ADDRESS hold time to /RD	0	
Trf	DATA float delay time to /RD		30



Picture 8. PCI-AI16HR의 Write timing diagram

Table8 . Definition of timing terms related to write timing

Symbol	Description	MIN[㎱]	MAX[㎱]
Tcw	/CS settling time to /WR	0	
Tdw	DATA setting time to /WR	30	
Twc	/CS hold time to /WR	0	
Taw	ADDRESS settling time to /WR	0	
Tww	/WR pulse width	50	
Twa	ADDRESS hold time to /WR	0	
Twd	DATA hold time to /WR	0	

5. Accessories (optional items)

5.1.Cable (dedicated cable for PCI-AI16HR)

When users use the terminal blocks shown in the table 10, the cables shown in the table 9 can be used and the picture 9 shows the shape of the cable.



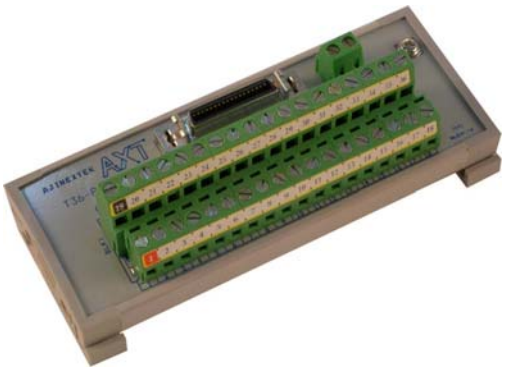
Picture 9. Cable C3736A-xTS

Table 9. C3736A-xTS specification (x = length)

Product Name	Current capacity	Number of pins	PITCH	Wiring form	Length(x)	Useable terminal blocks
C3736A-xTS	150mA	37pin - 36Pin	2.74-2.54mm (asymmetric)	Twist Pair	1m, 2m, 3m	T36-PR v1.0 T36-PR v2.0

5.2.Terminal Blocks

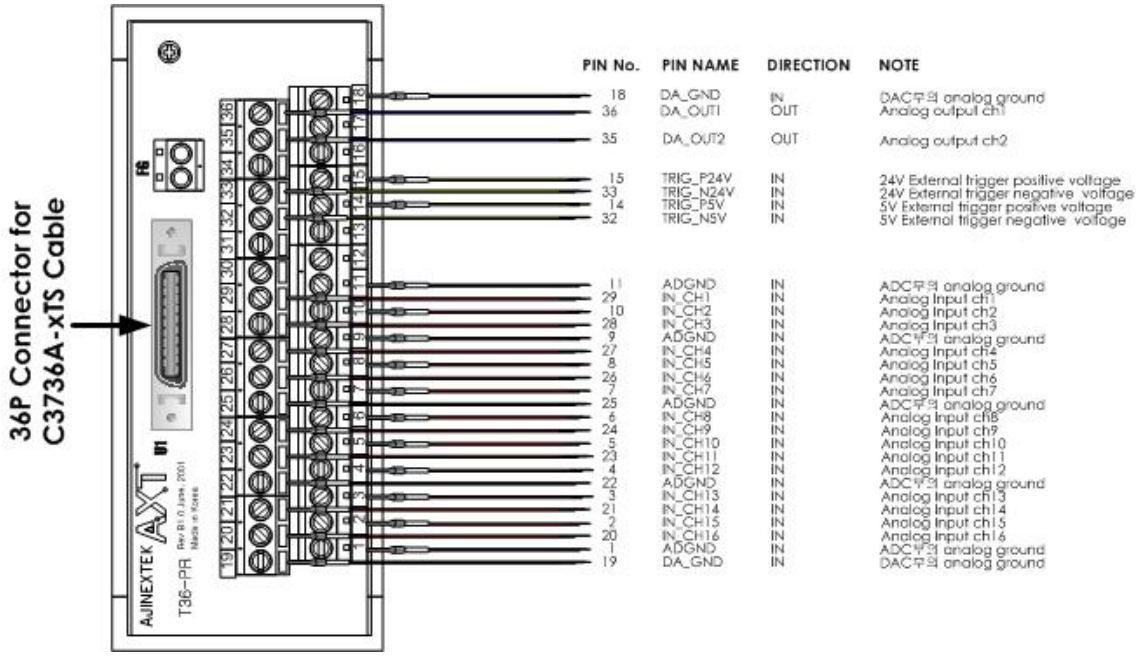
User can make and use user terminal blocks with reference to the pin array of the board connector. The basic terminal blocks provided by Ajinextek are designed to be used with the cable shown above, and users can choose terminal blocks as sown in the table 10 with the pin assignment as shown in the picture 10 and 11, and the specification is as shown in the table 10.



Picture 10. Picture of T36-PR v1.0 terminal block product

Table 10. Terminal block specification

Product Name	Height	Length	Width	Terminal itch	Fix Method	Terminal Type	Useable cable
T36-PR v1.0	43.0mm	200.0mm	52.0mm	5.0mm	Screw	Fixable	C3736A-xTS



Picture 11. T36-PR v1.0 Connection Diagram when used with PCI-AI16HR

Table 11. Description of T36-PR v1.0 & T36-PR v2.0 terminal blocks pin functions

Pin Number	Functions	Pin Number	Functions
1	Analog Voltage Input ch1	19	24V external trigger Positive voltage
2	Analog Voltage Input ch2	20	Analog Ground of ADC Part
3	Analog Voltage Input ch3	21	Analog Ground of ADC Part
4	Analog Voltage Input ch4	22	Analog Ground of ADC Part
5	Analog Voltage Input ch5	23	Analog Ground of ADC Part
6	Analog Voltage Input ch6	24	Analog Ground of ADC Part
7	Analog Voltage Input ch7	25	NC
8	Analog Voltage Input ch8	26	NC
9	Analog Voltage Input ch9	27	Analog Output ch2
10	Analog Voltage Input ch10	28	Analog Ground of DAC part
11	Analog Voltage Input ch11	29	Analog Output ch1
12	Analog Voltage Input ch12	30	Analog Ground of DAC part
13	Analog Voltage Input ch13	31	NC
14	Analog Voltage Input ch14	32	NC
15	Analog Voltage Input ch15	33	NC
16	Analog Voltage Input ch16	34	NC
17	NC	35	5V external trigger Negative voltage
18	24V external trigger Negative voltage	36	5V external trigger Positive voltage

5.3. Connecting Devices

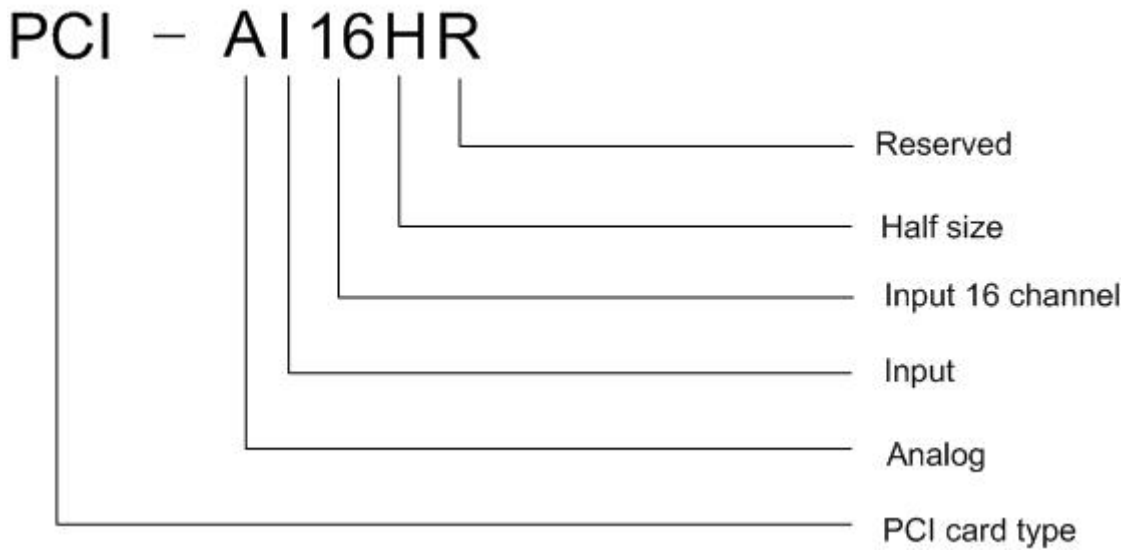


Picture 12. Connecting PCI-AI16HR board with peripheral devices (terminal board)

Picture 12 shows an example of the PCI-AI16HR board connected with peripheral devices including cable (C3736A-xTS) and terminal block T36PR.

6. Order Information (AnyBus Family)

Please refer to order information in the picture 13 to order products



Picture 13. AnyAIO model naming method

Table 12. Product family of AnyAIO modules

Product Name	Channel	Resolution	Direction	Internal DC-DC	Dimension
SIO-AO4RB	4	12-bit	Output	No	120mm * 45mm
SIO-AI4RB	4	12-bit	Input	No	120mm * 45mm
SIO-AO8HB	8	16-bit	Output	No	120mm * 92mm
PCI-AI16HR	16	16-bit	Input	No	120mm * 92mm
	2	16-bit	Output		

The contents of this manual can be changed without notice. The name of company, institution, production, people and event used in the examples are not real data. None of actual company, institution, product, people, or events are intended to be connected, and it shall not be interpreted that way. It is user's responsibility to comply with the corresponding copyright law. Separately from the right in the copyright, any part of this manual shall not be copied in any form or means (electronic, mechanical, paper copy, disk copy, or other method), shall not be coped for other purpose, and shall not be stored, introduced, or transmitted to the search system without explicit written approval of Ajinextek Co., Ltd.

Ajinextek Co., Ltd reserves all rights including patents, trademark right, copyright, or other intellectual property associated with the contents of this manual. Except the right explicitly given to you from Ajinextek Co., Ltd. by the written end user license agreement, providing this manual does not allow you to use any of patents, copyright, or other intellectual properties.